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THE INTERSECTION OF ORGANIZATIONAL IDENTITY, KNOWLEDGE, AND PRACTICE: ATTEMPTING STRATEGIC CHANGE VIA KNOWLEDGE GRAFTING

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We investigated the attempt of a high-technology R&D organization to transform into a market-oriented organization by “grafting” new, nontechnological knowledge. The intended strategic transformation did not succeed for reasons with wide implications for theory and research. Our findings suggest that the intersection of organizational identity, knowledge, and practice hindered the development of the new knowledge and undermined the broader strategic transformation effort itself. The failure of the graft revealed a previously underrecognized relationship between identity and knowledge that manifested itself in organization members’ efforts to preserve the collective practices that characterized how they used knowledge in accomplishing their work.

One of the great themes in the human experience is the idea of transformation. At the individual level, personal transformation has been the subject of countless inspirational and heroic stories and is a staple of social and literary lore. Transformation as a concept is so enduring and so engaging that it should not be surprising that it has leaped levels of conceptualization and become a recurrent theme in the study of one of humanity’s most prominent social creations, modern work organizations. Since the advent of organization study, some of the most captivating stories have been those that involved organizations transforming themselves into something very different from what they once were (Kanter, 1990; Pettigrew, 1985).

Many of these venerable stories have been accounts of successful transformation, perhaps because people prefer sagas of triumph out of ordeal. Yet the tendency to highlight success stories disguises the fact that most attempted transformations fail (Beer & Nohria, 2000; Sastry, 1997). Although most change is

difficult, genuine transformation is even more difficult, and the possibility of failure even higher. Failure, however, can be quite informative and instructive in revealing unrecognized relationships and in providing distinctive insights about some consequential organizational dynamics.

Through an examination of one high-tech organization’s failure to transform itself in the face of a discontinuous environmental shift, we provide insight into previously underexamined relationships among organizational identity, knowledge, and practice. Although organizational identity and knowledge represent two fundamental concepts in the study of organizing, their interrelationship has been notably understudied, with consideration kept strictly at a conceptual level (Brown & Starkey, 2000; Lant, 1999). Some evidence, however, suggests that a deeper understanding of how each affects the other, and the implications of this dynamic for other key organizational variables such as power and control, could provide valuable insight into basic questions of organizational adaptation and survival (cf. Cook & Yanow, 1993; Orlikowski, 2002).

A key context for studying the intersection of organizational identity and knowledge is found in attempts at strategic transformation. During transformation attempts, organizational identity is often destabilized and susceptible to change (Dutton & Dukerich, 1991; Gioia & Chittipeddi, 1991; Reger et

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al., 1998), current knowledge is often rendered inadequate and open to modification or replacement (Cook & Yanow, 1993; Glynn, Lant, & Milliken, 1994), and knowledge-identity relationships can become more apparent because of management's efforts to bring about the desired change.

The focal firm in our study, TekMar, Inc. (a pseudonym), chose to transform itself from an engineering-oriented ("technology-push") R&D organization into a business development-oriented ("market-pull") R&D organization by importing new business development and marketing knowledge from outside its boundaries, a process we refer to as "knowledge grafting" (cf. Huber, 1991). TekMar's traditional strength was basic research and development excellence in telecommunications technology; most of its members (including all of its top managers) were scientists or engineers who saw basic research and technological innovation both as the preeminent element of their work and the source of the company's distinctiveness. Facing a need to transform the firm from a protected R&D division of a large conglomerate into an independent, market-oriented organization, TekMar's leadership concluded that developing marketing expertise in-house was not viable and, therefore, chose to acquire it from the outside by hiring business development staff and integrating them into the upper echelons of the organization.

Organizations embarking on strategic change trajectories that deviate from their current strengths often face greater risks of failure (Haveman, 1992; Rumelt, 1995) than those that stick to what they know best. For that reason, exploration of processes that affect strategic inertia or adaptation have been the subject of recent works and calls for deeper understanding of strategic change processes (Burgelman, 2002; Siggelkow, 2002). Past research has shown that knowledge growth in organizations is usually path- and history-dependent—once internalized or institutionalized (Henderson & Clark, 1990), it is hard to change. Organizations are seen as engaging in search activities that build on their extant knowledge bases rather than reaching for more experimental, distal, or radical knowledge (Levinthal & March, 1993). These characteristics, however, pose a dilemma when strategic transformation requires developing new knowledge; the same expertise and skills from which the organization derived its competitive advantage in earlier times can become sources of rigidity (Leonard-Barton, 1992; Staw, Sandelands, & Dutton, 1981). This occurs, in part, because major change efforts produce tensions that bring issues of power and control to the fore. Thus, failed change efforts have been variously ascribed either to members resisting

change because they fear losing power (a more microlevel explanation) or to the existence of structural barriers that affect the acquisition and dissemination of new knowledge (a more macrolevel explanation). Our findings, however, suggest that TekMar's strategic transformation was unsuccessful because the change effort was profoundly affected by how knowledge and identity were embedded in collective practices—an insight that moves beyond simpler arguments based on resistance or barriers to change. Although TekMar's leadership successfully disseminated marketing knowledge throughout the organization, they were unable to bring about a much more fundamental change in how knowledge was actually used in the local, day-to-day practices of its members. Consequently, TekMar ended up jettisoning most of its imported business development expertise and falling back on its traditional strengths of basic research and development when the imported knowledge did not lead to changes in practice. We found that the ways in which TekMar's members used knowledge in their work practices recursively affected and were affected by their collective notion of who they were as an organization, and that power relationships became salient when top managers attempted to change their identity and/or their practices. More specifically, our study suggests a perspective unlike the predominantly cognitive focus found in previous work on organizational identity and strategic change. Here, it was identity's relationship with how knowledge manifested in organizational practices (a notably action-oriented concept) that was crucial. Thus, a focus on practice provides a fundamentally different way of understanding organizational knowledge, one that concentrates less on changes to knowledge content and more on changes to knowledge use (cf. Orlikowski, 2002). Practice, then, acts as a linchpin connecting organizational identity and knowledge. It is through practice that each influences the other—identity influences knowledge use by connecting knowledge to action, and knowledge use influences identity by providing the behavioral frames for its manifestation and maintenance—and thus becomes key to understanding the failure of TekMar's transformation attempt.¹

¹ In framing our introduction and the following conceptual overviews, we have employed the theoretical concepts that actually emerged from our study. In the grounded, interpretive research approach used here, the theoretical concepts and framework are grounded in and emerge from the data and analysis (Glaser & Strauss, 1967; Strauss & Corbin, 1990), rather than being derived

This study, then, is the story of a heroic failure, one that contains theoretical insights that have potential for reaching beyond the specific instance. We first review a set of literatures that provide a basis for considering the study's contributions, namely those relating to the intersection of organizational identity and knowledge. We then attempt to balance the phenomenological aspects of TekMar's story (the strategic transformation context) and the theoretical aspects (the interrelationship of organizational identity and knowledge use). Much of our data presentation focuses on the dynamics of TekMar's strategic transformation processes as a means of understanding deeper theoretical issues, which constitute the focus of our subsequent discussion.

ORGANIZATIONAL KNOWLEDGE AND IDENTITY

Research into organizational knowledge and research on organizational identity have proceeded along fairly independent tracks to date, although researchers have recognized the utility of linking them conceptually (Brown & Duguid, 2001; Kogut & Zander, 1996). In the following review, we discuss aspects of each literature that pertain to our study of TekMar's transformation attempt.

Organizational Knowledge

Although interest in knowledge is relatively new in organizational research and originally existed in the shadow of research on organizational learning, recent theoretical and empirical works have brought it to the fore (Grant, 1996; Kogut & Zander, 1992; Szulanski, 1996).

Traditionally, organizational knowledge researchers have looked at knowledge as a free-standing entity, focusing on the possession, creation, or

sharing of knowledge and explicating which types of knowledge relate to a particular phenomenon of interest (i.e., learning, change, innovation, etc.). For instance, research by Allen (1977; Allen, Tushman, & Lee, 1979) has provided key insights on the structural characteristics of formal and informal mechanisms that enhance or impede internal and external communication flows in R&D organizations. Likewise, numerous knowledge typologies exist; the best-known include explicit versus tacit (Nonaka & Takeuchi, 1995), simple versus complex (Zander & Kogut, 1995), and codified versus procedural (Kogut & Zander, 1992). The most representative organizational research from this perspective is that based on studies of knowledge transfer within and between organizations (Hansen, 1999; Szulanski, 1996), wherein the focus is on the objective and measurable characteristics of the knowledge being transferred (often characterized as "best practices" or simply "information").

A recent development in the literature on knowledge in organizations, based on the early philosophical works of American pragmatists such as Dewey (1938, 1949) and James (1963/1910), involves viewing knowledge less as an "object" and more as a dynamic phenomenon that manifests in the very act of knowing something. The move from objectified knowledge to processes of knowing shifts the focus to the specific activities that people engage in while solving problems. Dewey (1949), especially, argued against the existence of objective knowledge independent of interpersonal interactions; to Dewey, the act of knowing influences what is known. Thus, knowledge is fundamentally indeterminate and inheres in the specific activities that individuals engage in to deal with their day-to-day interactions with the world.

Working from the insights of pragmatist philosophy, Pentland (1992), Cook and Brown (1999), Brown and Duguid (2001), and Orlikowski (2002) further explored the links between knowledge and the social practices of organization members. The move from knowledge as an entity to knowledge-in-practice is a particularly important one because it is through practice that the value of knowledge becomes apparent. According to Cook and Brown (1999), practices refer to actions of organizational members *informed* by meanings grounded in their specific contexts. Thus, knowledge is viewed more as an ongoing dialogue between practice (action) and meanings (cognition). Merely holding knowledge about a technology or process does little good if that knowledge is not manifested in practice. Similarly, the same knowledge can be applied differently through practice, thus changing its role in an organizational milieu. The practice view of

from prior theory that guided data collection and analysis. A "purist" rendition of interpretive research reporting would, however, entail a lengthy and complex qualitative data presentation before the reader learns what the major theoretical dimensions and contributions are likely to be. In other words, the theory would normally appear *after* the data presentation (Daft, 1985). We suspend this interpretive reporting hallmark for the sake of advance clarity and employ the more traditional presentational strategy of providing a theoretical overview first, to preview the major findings and emergent model. It is important to keep in mind, however, that these concepts actually emerged from the study itself (along with consultations with relevant literature that were guided by the emerging thematic analysis).

knowledge also represents an important shift because focusing on the stocks and flows of knowledge (as a “possession” approach does) misses the manner in which knowledge use and value emerge “from the ongoing and situated actions of organizational members as they engage the world” (Orlikowski, 2002: 249). The key insight from this perspective that is pertinent for our argument is that organizational knowledge exists not as a body of canons independent of organization members. Rather, organizational knowledge becomes an *organizational* concept through its local manifestations in the practices and activities of organization members.

Organizational Identity

Research on organizational identity has flourished in the two decades since Albert and Whetten (1985) first defined the concept as those features of an organization that its members deem to be the most central, distinctive, and enduring. Recent research has taken their original concept in many directions, so that there are now varied perspectives on how best to conceptualize identity. Responding to Corley, Harquail, Pratt, Glynn, Fiol, and Hatch's (2006) recent call for researchers to state explicitly how they are conceptualizing organizational identity, we take a social constructionist view that organizational identity entails members' consensual understanding of “who we are as an organization.” In this view, identity is less enduring and distinctive than it is in Albert and Whetten's (1985) definition, but it is no less crucial to organizational survival and growth (Gioia, Schultz, & Corley, 2000; Glynn, 2000; Golden-Biddle & Rao, 1997).

Two aspects of recent organizational identity research are particularly relevant to this study: its relationship with strategic change and previous research's implicit bias toward cognitive aspects of identity. First, studies on organizational identity and strategic change have highlighted the problems organizations confront in changing their identities when faced with new imperatives (Bartunek, 1984; Gioia & Thomas, 1996). Identity has often been depicted as a relatively enduring organizational feature that impedes strategic change, mainly because members are viewed as avoiding learning to preserve existing conceptions of themselves and their organizations (Brown & Starkey, 2000). Although organizational identity's role in strategic transformation is likely to vary over change contexts, it is clear that identity is likely to influence any attempt at transformative strategic change. This is because organizational resources, especially

knowledge, skills, and expertise, are likely to be influenced by the basic assumptions and frames of reference that organization members use to define “who we are” as an organization (Kogut & Zander, 1996; Oliver, 1997). Even as early as 1957, Selznick suggested that an organization's distinctive competence was closely aligned with members' collective sense of self-definition. It is still unclear, however, how organizational identity interrelates with processes critical to strategic transformation.

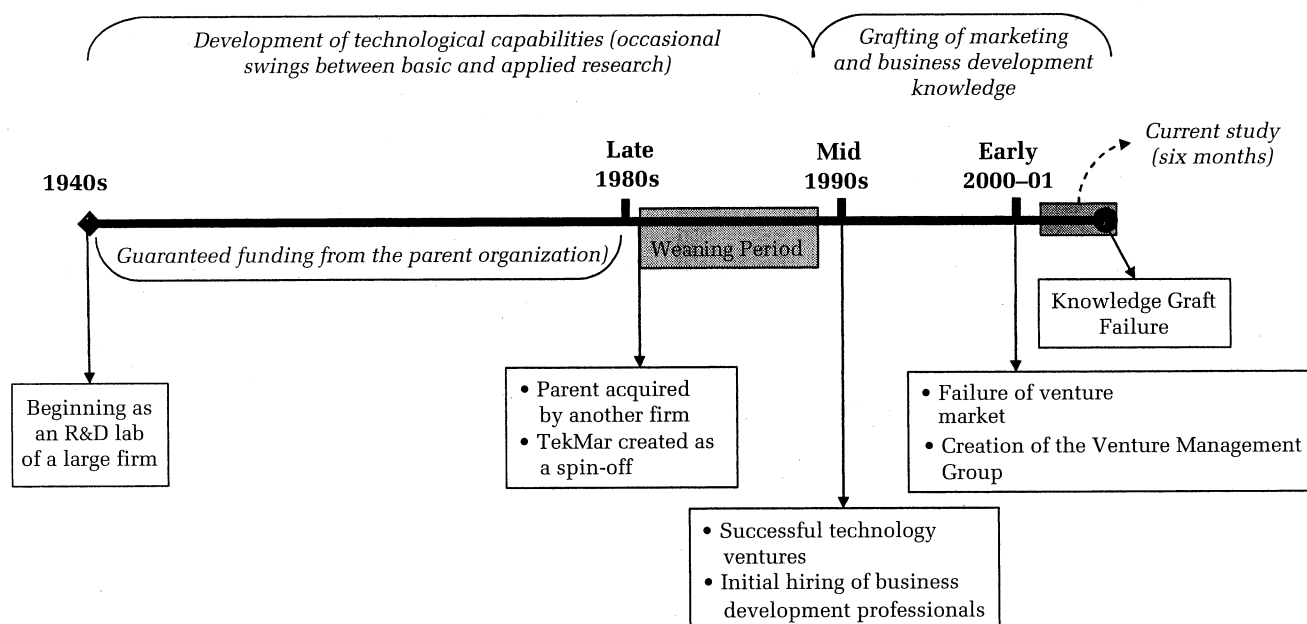
The second salient aspect of existing identity research is the predominantly cognitive approach taken to the study of organizational identity during strategic change (Bartunek, 1984; Gioia & Chittipeddi, 1991), whereby identity is treated as something residing principally in the minds of organization members. Any exploration of organizational identity's connections to strategic change, however, implies broader linkages. Dutton and Dukerich (1991), for instance, found that organization members' sense of identity was associated with the organizational routines, skills, and decision-making processes undergoing the change. Similarly, Corley and Gioia (2004) found that changes in organizational identity sparked by an organizational jolt corresponded with behavioral changes among the organization's members, especially those in leadership positions. Scholars' current conceptions of organizational identity, therefore, need to transcend purely cognitive views by accounting for identity's relationship with the collective practices that characterize how organization members conduct their daily work.

With little prior empirical research to draw on as a base, researchers have found it a challenge to empirically examine the relationship between a dynamic conception of organizational knowledge and a more action-oriented conception of organizational identity. Viewing organizational identity and knowledge as embedded in the social practices of everyday work life could, however, provide a strong foundation from which to explore this relationship and gain insights into the key elements for understanding not only strategic transformation, but also, more importantly, the essence of organizational identity and knowledge in modern organizations.

RESEARCH CONTEXT

The findings of this study are best understood in the context of TekMar's historical evolution, the backdrop for the firm's more recent attempts to graft on new business development and marketing knowledge. Figure 1 illustrates the timeline of Tek-

FIGURE 1
TekMar Timeline^a



^a Not to scale.

Mar's organizational evolution and marks the significant events that form the focus of this study.

TekMar's Historical Roots: Science for Science's Sake

TekMar, Inc., began as a dedicated research lab of a large American electronics firm in the early 1940s, flush with funds to do scientific research. During its time as an R&D division, the organization's human resources largely comprised talented scientists and engineers. In these early years, the R&D division's members carried out basic and applied scientific research and developed innovations in the arenas where its parent company competed. The next distinct period in the company's history, spanning the 1950s and 60s, was a time when basic research gained clear ascendancy over applied research. During this period, the R&D division ventured into uncharted scientific areas that broadened its knowledge base and allowed its scientists to create new fields with their discoveries. The subsequent period, the early 1970s, however, saw a return to the application of technologies to solve manufacturing issues among the other divisions of the parent corporation. Despite these "pendulum swings" between basic and applied research, the R&D division continued to be technology-focused, with its research agenda defined either by the technology requirements of the parent company or by top management's visions of

the future. It was also during this period that the division's scientists achieved significant technological and scientific breakthroughs (and national recognition) that helped reaffirm their status as top-notch scientists in their respective fields. Our informants consistently used the term "technology push" to label the dominant mind-set of the organization. A pure technology-push mentality is characterized by internal creative processes focused on designing and developing cutting-edge technology (often without an obvious commercial application); this technology is then pushed into the marketplace regardless of current market interest.

In the mid 80s, the parent company was acquired by another large technology conglomerate, after which the R&D division was eventually reconstituted as a freestanding firm wholly responsible for its own strategic direction and performance. The 90s was a weaning period during which the new organization, TekMar, Inc., had access to initial (albeit progressively decreasing) support funds that would help it through the first five years of its independence. Subsequent to this weaning period, TekMar would be open to market forces and responsible for seeking its own revenues and funds for research. The newly found independent status was a watershed in TekMar's history and a major jolt for the organization. All of a sudden, the same scientists and engineers whose main jobs for years had been to work on projects that excited their intellects now had to be responsive to market de-

mands because TekMar's well-being depended on such responsiveness. Perhaps most tellingly, now TekMar's leadership would only approve exploratory funding if a proposed project was deemed to have commercial potential (yet no formal sense of what "commercial potential" meant existed in the organization). TekMar's engineer/scientists tried some idiosyncratic, even serendipitous business development practices during this time, but there was no concerted organization-wide business development effort. In any case, science for science's sake was no longer a viable option.

Knowledge Grafting at TekMar: Science for Survival's Sake

As a facet of completing their forced independence, TekMar's leadership began a process intended to make it a more market-driven company focused on new ventures. To capitalize on the burgeoning opportunities in the 1990s' venture market, top management decided initially to infuse business development expertise into the organization by bringing in new, nontechnical employees with business development backgrounds to join project teams, as a quick way to make the organization "market-oriented." The process was not an easy one, as the business development people faced opposition because of their ill-defined roles and their inability to understand the technological intricacies involved in the daily work of the scientists and researchers. The turnover rate for the business development professionals was quite high. In the late 1990s, TekMar was reorganized into several business units, each focusing on a specific technological area.² The reorganization was meant to diversify the technological base of the company and streamline management of its operations. Reorganization also provided an opportunity for the company to try its hand at creating ventures around new technology, an experiment that found some localized success in the cash-rich venture markets of the late 90s and early 00s.

TekMar's initial successes in ventures spurred the company toward what many of our informants called "venture fever." The initial successes spawned a spate of new ventures, which the company's leaders hoped would generate much-needed financial growth. A few of these ventures succeeded at the business-unit level, but none generated the kind of profitability nec-

essary to sustain TekMar at the organizational level. Then, early in the new millennium, TekMar suffered another environmental jolt when the technology bubble burst and external funding for tech ventures dried up. What once looked like a promising avenue for growth turned into one fraught with failures. As a senior vice president said,

In the market space we went from market exuberance to market pessimism, so people were flying high and thought they were going to go public, even just six months ago. Now people are worried about how long the cash is going to last.

TekMar's leadership responded to this jolt by reinforcing the drive toward an even greater market-pull orientation with a more aggressive knowledge-grafting strategy. It created a new executive-level group, the Venture Management Group, to further expedite the development of critical venture management and commercial applications expertise.³ The members of this new group were experienced market professionals with strong skills in venture development and funding. Their presence, however, and especially the fact that they reported directly to the CEO, engendered tension and conflict about whether the engineer/scientists or the Venture Management Group had the power to control the way work was done within TekMar. The net effect was that power was transferred from an interpersonal level (between scientists and business-unit managers) to a structural level (between business units and the Venture Management Group), which meant that the latter was now able to tell engineers/scientists how to do their work. In particular, the Venture Management Group imposed formal practices for sharing knowledge and reporting business development progress, which led to ambivalence at TekMar because, although the scientists readily accepted the need for new business development knowledge, they were less sanguine about the new work practices.

As shown in Figure 1, our research team gained access to TekMar soon after the introduction of the Venture Management Group, when we were granted permission to interact with executive and middle-level managers attempting to implement the formal knowledge graft. Among the many apparent inconsistencies that surfaced at TekMar, the key one on which we focus is how an organization essentially made up of brilliant scientists and en-

² TekMar was restructured into nine business units, and the organization as a whole had about 750 members when we gained research access.

³ The Venture Management Group comprised three members, including its head. All members of the group had executive power to control project funding for all business units.

gineers (even in its management ranks) found it so difficult to incorporate the new knowledge that was aimed at making their research initiatives more market-oriented. This state of affairs is, in principle, common to many high-tech firms that hire mainly for technical expertise and then find themselves confronting environments that demand greater business development acumen.

RESEARCH APPROACH AND METHODS

Using an exploratory case-study-based research design (Eisenhardt, 1989; Yin, 2003), we adopted a grounded, interpretive approach for the purpose of explorative richness, as little theoretical precedent exists for deductive inquiry in this domain. An interpretive approach gives voice in the interpretation of events to the people experiencing them, and “the native’s” point of view is an important constituent of the analysis (Van Maanen, 1988). An interpretive approach does not imply suspension of a researcher’s judgment, however. Although interpretive work affords the voices of informants an important role, we as researchers assume the task of further interpreting and structuring the interpretations of the respondents in light of both contextual factors and prior theorizing (Strauss & Corbin, 1990) to develop a final emergent model.

Research Procedures and Data Sources

Interviews. To obtain multiple perspectives on the strategic change effort, over a period of six months we conducted 34 in-depth interviews with eight senior executives: the CEO, the chief operating officer (COO), senior vice presidents, and the head of the Venture Management Group—individuals together comprising the entire senior leadership team—and 12 midlevel managers (almost all the business-unit heads, their corresponding business development managers, and the two other members of the Venture Management Group’s personnel). Our informants’ tenure in the organization ranged from six weeks to 36 years. They comprised all the organization members who either planned or were responsible for implementing the strategic change effort. All the informants, excepting the business development professionals, were engineers or scientists. The midlevel managers, who were also engineers or scientists, served as the voice of the employees who reported directly to them. For these reasons, we were confident that the interviews represented the wider organizational membership. We also interviewed several informants after the failure of the graft.

The interviews varied in duration, but all fell in

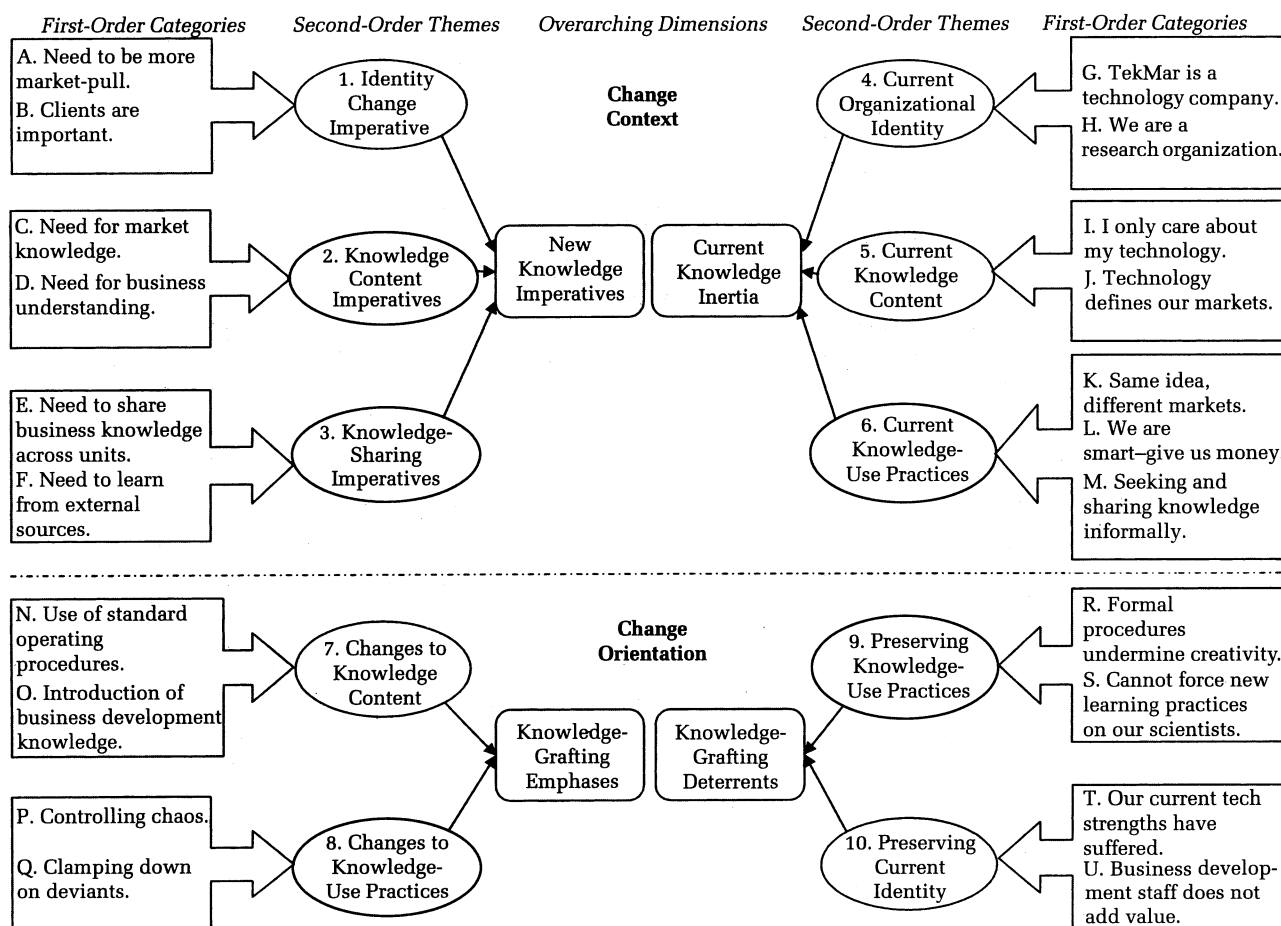
the range of 60–120 minutes each (all informants’ initial interviews lasted at least 60 minutes), and all but four impromptu interviews were recorded and transcribed verbatim. Initial interviews involved questions about tenure with the company, background on the organization’s structure and culture, perceptions of its historical evolution and current situation, and involvement in its strategic processes. Secondary and tertiary interviews became progressively more structured as themes emerged. Questions during these interviews involved areas such as perceptions and actions related to strategic change within the company, relationships within and between business units, how knowledge creation and sharing occurred, and the progress of ongoing strategic issues. We urged all informants to take the role of spokesperson for the organization or for his/her business unit, depending on the question, in a concerted effort to keep the focus on organizational level issues and gain more detailed insight into the collective nature of TekMar’s strategic change efforts. Our research team followed a “24-hour rule” for first writing down individual notes and thoughts about each interview and then discussing them as a group. We also audio-taped and transcribed all post hoc research team discussions to keep a running record of observations, informant interpretations, researcher insights, and further questions. We used the observations, interpretations, and insights contained in these field notes and the notes of research team discussions to supplement the transcribed interviews, to recursively understand the emergent findings, and to modify the interview protocols for subsequent interviews.

Observation and archival sources. In addition to gathering the interview data, we observed and took notes at several top management strategy sessions and meetings of the Venture Management Group (audio-taping was not allowed). Once again, we discussed our notes immediately after each meeting and used these notes during data analysis and in the modification of future interviews. We also analyzed archival data in the form of company Intranet pages and Internet Web sites, published news articles, internal company documentation, and internal company presentations by senior management to obtain historical reference points for the current dynamics of strategic change in the organization.

Analytical Approach

As we collected the data, we also inductively analyzed it, adhering closely to the guidelines specified for naturalistic inquiry (Lincoln & Guba, 1985)

FIGURE 2
Data Structure



and constant comparison techniques (Glaser & Strauss, 1967; Strauss & Corbin, 1990). These approaches provided the basis for rigorous collection and analysis of qualitative data and helped us determine the sampling and content foci of later data collection. Additionally, they provided the basis for delineating themes and aggregate dimensions through the examination and comparison of key events (Isabella, 1990).

In our initial rounds, we coded each interview separately on the basis of “in vivo” words, phrases, terms, or labels offered by the informants; that is, we discerned first-order codes (cf. Van Maanen, 1979). We then reread each interview several times, each time marking phrases and passages that were similar to and different from each other, to discern similarities and differences among informants. We relied on constant comparison of multiple informants and over time to detect conceptual patterns (Glaser & Strauss, 1967). To systematize the data coding, we used a computer-based qualitative analysis program, NVivo 2, that enabled us to record and cross-reference the codes that emerged from

the data. After going through multiple interviews, we began discerning codes across informants that were similar in their essence. We collapsed these codes into first-order categories, employing language used by the informants that expressed similar ideas. We continued coding interviews in this manner until we could not ascertain any more distinct conceptual patterns shared by the informants.

Along with developing first-order categories, we started discerning links among these categories. These emergent links enabled us to collapse first-order categories and cluster them into theoretically distinct groupings, or second-order themes (researcher-induced concepts, cast at a more abstract level, albeit using informant labels if they represented emerging theoretical concepts). We then assembled the second-order themes into overarching dimensions that enabled us to finalize a theoretical framework that linked the various phenomena that emerged from the data.

Additionally, to ensure confidence in the assignment of codes to categories, we employed two other coders to independently assess our coding

scheme.⁴ Overall agreement was strong; those instances in which there were disagreements, either between assessors or between us and an assessor, formed the basis for discussion and modification of the coding scheme until consensus was achieved. This extra step provided an independent perspective on the trustworthiness (Lincoln & Guba, 1985) of our coding scheme and the emergent theoretical framework. Additional steps taken to help ensure the trustworthiness of our interpretations included explicitly distinguishing between first- and second-order data (Van Maanen, 1979) in frameworks and reports and conducting “member checks” with our informants to gain confidence that the interpretive scheme was sensible to, and affirmed by, those living the phenomena of interest.

EMERGENT FINDINGS

This study of attempted strategic transformation via knowledge grafting involved a number of complex phenomena, but the intersection of organizational identity and organizational knowledge use in practice was pivotal in understanding the events we observed. Figure 2 depicts the emergent data structure. The phrases in the boxes on the left and right sides of the figure show the first-order categories (terms adequate at the level of meaning of the informants); the concepts contained in the ovals show the assembly of these first-order concepts into second-order analytical/theoretical themes; and the four boxes in the center show the overarching dimensions that emerged from the analysis. Table 1 contains additional supporting evidence, keyed to Figure 2, showing representative infor-

mant (first-order) quotes underlying the second-order themes.

Although we were concerned primarily with these main analytical categories, themes, and dimensions, it soon became apparent that two larger domains subsumed them. We labeled these domains the “change context,” to describe the milieu within which the strategic change (via knowledge grafting) initiative took place, and “change orientation,” to describe the key change and grafting issues that were most implicated in the eventual failure of the transformational change effort. Each domain has two countervailing dimensions that represent the essential tensions (sometimes characterized as power or control concerns) that the informant interviews made evident. The first domain denotes a tension between *new knowledge imperatives* (upper left quadrant) arising from a changed environment (which was perceived by TekMar’s senior leadership as demanding acquisition of business development knowledge) and *current knowledge inertia* (upper right quadrant) arising from TekMar’s members’ identity and embedded work practices. This tension played out during the attempt to implement change (lower half of Figure 2) in terms of the *knowledge-grafting emphases* touted by the Venture Management Group and the countervailing *knowledge-grafting deterrents* that occurred because of conflict between the orientation of the Venture Management Group and the rest of the organization about the appropriate approach to the graft.

Figure 2 is important for analytical reasons, because it shows the initial emergent data structure (first-order concepts and how they relate to second-order themes and the overarching dimensions). It represents, however, a rather static picture of the operating processes involved. To understand what happened in the case in dynamic terms, we conducted further analyses of the key themes and derived Figure 3, which in one sense is Figure 2 “set in motion.” Figure 3 represents a synthesis that shows the chronology of the grafting experience and connects the emergent analytical concepts together. It therefore more clearly captures the dynamism of the change process (for example, static concepts like “imperatives” become dynamic concepts like “initiatives”).

In Figure 3, we focus at the theme level, which is more informative than focusing on the coarser-grained dimensional level. As will become evident, the tensions portrayed in the overarching dimensions ultimately turned out to be manifestations of deeper issues and processes better captured by

⁴ Although interpretive research would not normally include assessments of intercoder agreement, we nonetheless were curious to see if we had convergence on key aspects of our coding scheme. A high degree of agreement would not only lend rigor to our analyses, but also bolster our confidence in the plausibility of our interpretations. We therefore engaged two researchers who had no prior connection with the research site. We used a modified Q-Sort approach (Stephenson, 1953) in which we asked these coders to assign several informant quotes/phrases into 21 first-order categories. We gave each coder brief definitions of the second-order themes and explained and clarified the meanings of individual categories as requested by the coder. We then asked the coders to assign the 21 categories to ten second-order themes. We calculated the agreement level of each assessor with our coding scheme to determine intercoder agreement. The average intercoder agreement was 88 percent, signifying a high level of agreement concerning our coding scheme.

TABLE 1
Dimensions, Themes, Categories, and Quotations

Second-Order Themes and First-Order Categories	Representative Quotations
Overarching dimension: New knowledge imperatives	
1. <i>Identify change imperative</i>	
A. Need to be more market-pull.	A1. That's the difference between technology push and market pull. For the most part, training of everybody on the technical and for a good part of the managing directors is technical push. That's their training. What we are learning is market pull. (manager, planning) A2. We're trying to create an organization of marketers. (business development manager 1)
B. Clients are important.	B1. We need to listen to clients. (CEO presentation to all employees) B2. We need to extend beyond solving problems that clients tell us. We need to go out when our phone doesn't ring. How else would I know if the clients just aren't calling me or the phone doesn't work? So you should get out there. (business development manager 2)
2. <i>Knowledge content imperatives</i>	
C. Need for customer and market knowledge.	C1. What I want is my group head to go out with technology in his back pocket, but listen to a client and say, "What do you need"? And then, "What have I got?" I want them to uniquely understand the sales process. (business-unit manager 1) C2. I would say in many areas we are consciously incompetent, leaning towards consciously competent. This whole area of market intelligence and understanding the world, rather than waiting for a client to say, "Give me this." We are kind of sitting around waiting for somebody to say, "Give me this." (senior vice president 4)
D. Need for business understanding.	D1. You really have to be more business focused and [it] really comes down to your numbers and your financials. (business development manager 3) D2. What they are being asked is not how good is the technology, but where are you going to use the technology, and who wants that technology, and why even if you had it would it replace what exists? So where in the marketplace is the pull coming from? (manager, planning)
3. <i>Knowledge-sharing imperatives</i>	
E. Need for sharing business knowledge across units.	E1. What will typically happen is what I see over and over. This person will hire somebody and say okay, "go make me money" and what that means is go do the learning curve and learn every market. Go figure out where you can make money. The learning curve is way too much and it's a recipe for disaster, because no one individual [in each business unit] can go through times X number of markets. (business development manager 3) E2. I saw a great meeting where the BD guys came together at one point, where they said "you know in order to really great proposals around here, we really need to know what's going on in the other business units." (Venture Management Group head)
F. Need to learn from external sources.	F1. When the issue becomes understanding marketplaces and figuring out whether it's a business or not. As opposed to just an interesting technology; you are solving a technological problem. That's when you have to get out. So it gets to a certain point where you say I can't learn anymore inside the building. (Venture Management Group member 1) F2. I think the thing that I would take is the focus on the market, as opposed to the focus on the technology. It's an outward focus as opposed to an inward focus. (business-unit manager 1)
Overarching dimension: Current capability inertia	
4. <i>Current organizational identity</i>	
G. TekMar is a technology company.	G1. The whole idea of technology is what we focus on. We're a technology company. I just believe the reason why we're structured this way is because it's a technology company. (business development manager 1) G2. We had a great idea for a company and went out in the marketplace and did all that, but we brought low technology to it. The VCs said, "Thanks for the idea. What do you bring to it?" So eventually the technology has its role. The thing is, you don't want to lead with technology, and you don't want technology to be the only thing that drives you, but technology is really the foundation. (senior vice president 1)

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TABLE 1
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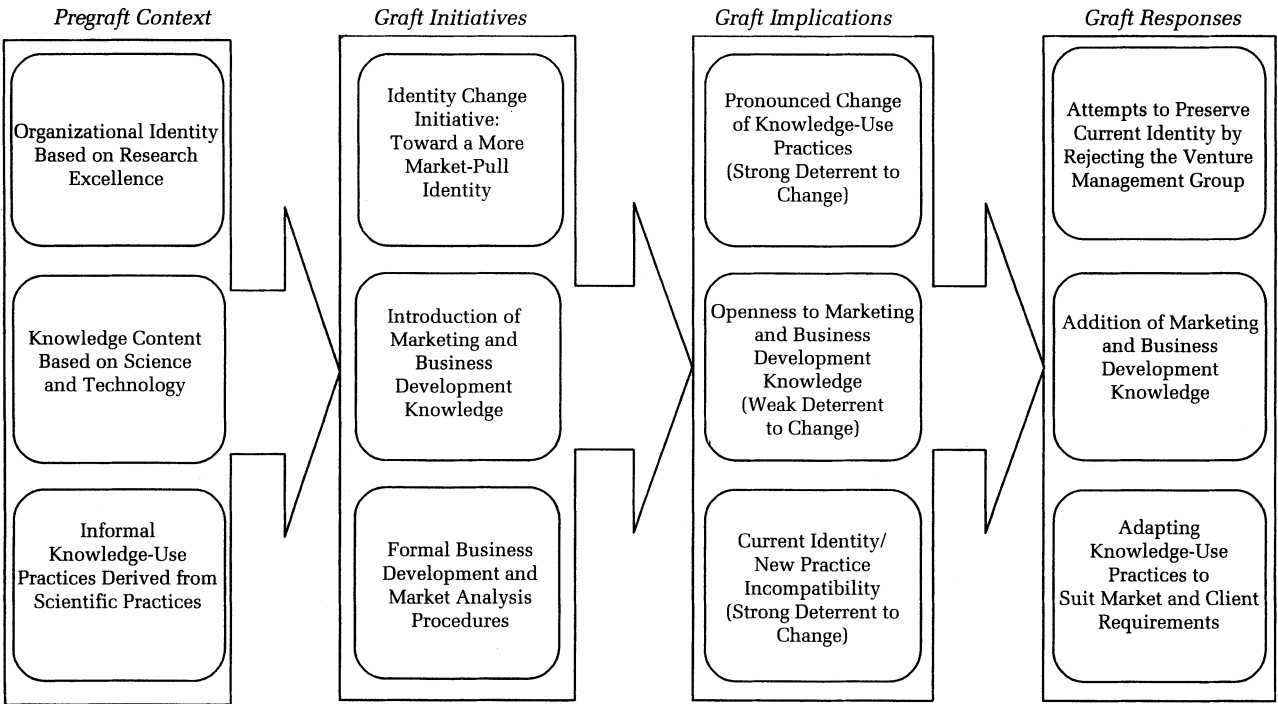
H. We are a research organization.	H2. The whole organization is managed by researchers. (business development manager 3) H3. I think it's an identity. What kind of a company are we? We were on a path for the past few years of having a couple hundred million dollars in value. We were thinking we were going to be fully endowed and this model is great. This is where the venture IPO market dried up . . . so we have got to focus on the basic core, you know, contract research. (Venture Management Group member 2)
5. Current knowledge content	
I. I only care about my technology.	I1. Whatever someone else does in the market—you could say consumer electronics in the market (I don't know if it is or isn't)—but what anyone else does in that area doesn't concern me. I only care about my technology. (business development manager 2) I2. For most employees and a good part of senior managers, the training is essentially technical. (manager, planning)
J. Technology defines our markets.	J1. We haven't carved the company into market spaces. We're technology carved [rather] than market. (business-unit manager 4) J2. There are a lot of things that we're very good at here. But that's technology, it's not business necessarily. (business development manager 3)
6. Current knowledge-use practices	
K. Same idea, different markets.	K1. "Z" is an interesting application of technology from metal "A" to metal "B." A little bit of black art. How do you get the two metals to expand at the same rate? (manager, planning) K2. But the question was, how do you get this technology out in the market? How do you convince a customer to put you know \$10 million into it. So it had to be linked to some high-value production application. I had been doing some stuff in optical components before that trying to sell something and it occurred to me that this technology could be quite relevant in this new market space. So once we put that story together, all of a sudden we had a venture idea out of it. And when we added potential for revenue and profits, it started to look good. (business-unit manager 4)
L. We're smart—give us money.	L1. So it's quite interesting when we try to develop new business. One very common one is that we take the capabilities in which we have 20–30 years of experience and we know the industry, know the shortcomings of the current technologies and we come up with a new concept for the market. (business-unit manager 3) L2. In many cases you'll find that we can apply the right technical resources [scientists]. We can solve technical problems and then we can show the client how it's good for him. That's how we usually get our business. (business-unit manager 2)
M. Seeking and sharing knowledge informally.	M1. We would do them [seek help] ourselves independently. Like with this optical project, I would eventually occasionally say, "Maybe I'm really wrong." So I would go either talk individually to an expert or I would call a small meeting of two or three experts. (senior vice president 1) M2. I think in all cases it happens informally. Like let's talk about our ventures again. We've got only a handful of executives that serve on the boards of the ventures, right? So if the same guy is sitting on two boards, and if he knew the lessons learned from one, you would think at the boards' meetings he would kind of transfer that knowledge, those lessons learned, to the other. (senior vice president 4)
Overarching dimension: Knowledge-grafting emphases	
7. Changes to knowledge content	
N. Use of standard operating procedures.	N1. What you would be working from was a template that has been refined and set out, and said, "Okay, fill in this template." Here are the four questions. (manager, planning) N2. We made them (business units) do our standard quad chart; you know, what's the need? What's the advantage? What's the benefit? Who's the competition? Why should I care about this? How much money do you want? (Venture Management Group head)

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TABLE 1
Continued

O. Introduction of business development knowledge.	O1. We ask them questions like, do you understand your distribution channel, do you understand your market? If you don't you ought to go and get to talk to somebody who has had some experience in it. (Venture Management Group head) O2. The COO has absorbed the lesson and is a leading proponent of it. The senior management understands it fully, so as they [organization members] interact and try to get things done, they are critiqued by the senior staff basically. So it spreads down to about the business-unit directors because they are the ones that have the most influence on the overall direction of the business unit. (manager, planning)
8. <i>Changes to knowledge-use practices</i>	
P. Controlling the chaos.	P1. There is a fear in this organization of putting too much rigid structure around creative people. In fact I believe the exact opposite is true. That if you give people a box in which to think, they are going to make the maximum out of that box. But if you let them think around the whole world, it might not come together quite as readily. So I believe this is a fundamental flaw here. I think the controlled chaos keeps us from maximizing our potential. (senior vice president, human resources) P2. I've heard this term used even by executives, chaos, and put some other adjectives around it. Somewhat controlled. Genius chaos. And I think you know some people that are in high positions sort of believe that's the way to really create value. (business development manager 1)
Q. Clamping down on deviants.	Q1. We really clamp down hard the minute we find out the guys have been talking to venture capitalists and it hasn't gone through my organization (venture group). Who are you talking to and why? (Venture Management Group head) Q2. Once it gets put on one of these official lists, then you got to keep reporting on it. You got to report on the progress, what you have done, what you haven't, and there's enough on that list to let people know what the idea factory hasn't got down. (Venture Management Group member 2)
Overarching dimension: Knowledge-grafting deterrents	
9. <i>Preserving knowledge use practices</i>	
R. Formal procedures undermine creative chaos here.	R1. I don't know what kind of a SOP [standard operating procedure], if there should be an SOP in this building. That's why I think we have freedom. (COO) R2. It's a wild west kind of thing. You've got to kind of manage the chaos. But if you formalize the chaos so that you eliminate the chaos, it won't work. (Venture Management group member 2)
S. Cannot force new learning practices on our scientists.	S1. We have all this stuff out on the Web. We have broadcast e-mail that we use all the time. We have this all-hands talk all the time. So we use every technique that I mean most companies wouldn't even begin doing half of the stuff we use and yet people—just people are not paying attention. They don't because they're scientists. They don't see the relevance of reaching outside of their own limited mind-set. (senior vice president, human resources) S2. I tell you this place, if you try to put requirements to come back with a trip report . . . you know, people are going out on their own time to try to learn for themselves, to share knowledge, other than anecdotally . . . my gut reaction, and I guess I've only experienced it and test-marketed to a small degree, is that imposing reporting things like that just does not fly here. (Venture Management Group head)
10. <i>Preserving current identity</i>	
T. Our current technical strengths have suffered.	T1. I think if we looked at the staff that was here [a few years ago] . . . I think there was a larger fraction of people interested in research that were, you know, high-class researchers. Now, I'd say, if anything, what concerns me is that the people who are strong in technical research have thinned out quite a bit. (business-unit manager 5) T2. I think in the new format, we became more centralized. There is a tendency to put a structure around [a product champion], making sure he got his act together. We doubled the tendency to improve things. What that did was take away the initiative of the champion. (COO)
U. Business development professionals are not adding value.	U1. As I said, the jury is out as to whether or not that was the wisest thing to do [creating the Venture Management Group]. Maybe we should have done it in a steadier, slower process. (senior vice president 1) U2. If you go ask individuals on a one-on-one basis, "Should we not do ventures anymore?" They will resoundingly say, "We want to have ventures!" Are the marketing professionals really just an oversight group, or are they truly adding value and championing the project, so that's where it gets a little murky. (Venture Management Group member 2)

FIGURE 3
Key Dynamics Underlying Deterrents to Strategic Change via Knowledge Grafting at TekMar



these themes.⁵ This approach to reporting our findings also allows us to demonstrate the close connections among method, data, and (grounded) theory. Figure 2 provides the bridge between our method and our data; Figure 3 in turn provides the essential bridge between our data and the emergent grounded theory (discussed in the section following the findings). For these reasons, we base the following findings narrative on Figure 3.

As shown in the first column of Figure 3, the character of TekMar’s current organizational identity (based on research excellence), knowledge con-

tent (based on science and technology), and knowledge-use practices (also science-based) jointly constituted the *pregraft context* into which new *graft initiatives* (second column) were introduced in the form of attempts to change TekMar into a market-driven organization that used formal business development procedures for knowledge use, sharing, and reporting. These initiatives entailed some serious *graft implications* (third column) for knowledge-use practices and for existing identity that led ultimately to a number of adaptive *graft responses* (fourth column).

I. Pregraft Context

The most prevalent theme in the interviews and secondary data sources relevant to the pregraft context (column 1, Figure 3) was TekMar’s historical *organizational identity*.⁶ “We are a technology organization!” was perhaps *the* most common informant statement. Even the senior leaders of the change effort alluded to TekMar as predominantly a technology research organization. As one senior vice president noted:

The core asset for the company, and its real reason for being, is the same as it ever was—mainly to have

⁵ Although Figures 2 and 3 have different formats, their constituent themes correspond closely. One important difference is that further analysis of several of the themes in Figure 2 generated a finer-grained parsing into two additional themes in Figure 3. For instance, in further analyzing the “changes to knowledge content” and “changes to knowledge use practice” themes of Figure 2, it became apparent that the outcomes of the imposed changes to knowledge content and knowledge use led to a dysfunctional incompatibility between existing identity and the changes in work practices (as reflected in the inclusion of this theme in the third column of Figure 3). A related analytical rationale explains the additional theme in the fourth column of Figure 3, which emerged after we reanalyzed the “preserving knowledge use practice” and preserving current identity” themes from Figure 2.

⁶ The italicized in-text terms represent the main second-order themes and are keyed to the cells in Figure 3.

in place a very broad-based set of researchers who, especially in the top ranks, are really quite world class as far as technology goes.

The notion that TekMar was first and foremost a technology company was even more ingrained among the middle managers, all of whom came from the scientist-engineer ranks. This belief reflected their own backgrounds and the beliefs of the scientists and engineers they managed. Scientific and engineering brilliance was a hallmark of the company and a point of pride to its members. Overall, these engineers/scientists unequivocally saw themselves as members of a *TekMar* research community; one where a clear technology-based TekMar identity prevailed.

In line with this identity, TekMar's existing *knowledge content* was also predominantly scientific and technological—even when it came to being a business. Time and again, both senior and middle-level managers identified core businesses, markets, and processes along technological dimensions, which was quite consistent with TekMar's history as a technology-push organization. The senior vice president of human resources, when asked to comment on this tendency, remarked, "They tend to still want to put all the emphasis on science and technology, and the scientists really have a tough time appreciating that some of our best ventures actually have very little to do with science." We should note, however, that in the early period of TekMar's independence, there was widespread recognition that marketing and business development knowledge would now need to have a more prominent place in organization members' thinking about applying technology and turning innovations into commercial ventures.

Our informants characterized their *knowledge-use practices* as informal, atomistic, ad hoc, and even serendipitous. As one business-unit manager noted, "Most of it happens informally; people getting together when they need help, etc. . . . often with a bunch of scientists sitting at the same table for lunch." These practices enabled groups to draw simultaneously on different forms of knowledge possessed by different people. For example, the COO, an accomplished scientist himself, summarized this stance as, "There are no absolute truths, no standardized 'right way' of doing things. Let intelligent people pick and choose what they want." The emphasis on the qualifier "intelligent" exhibited taken-for-granted beliefs about TekMar's members being world-class researchers and what this meant for how they worked, providing further evidence that TekMar's organizational identity

strongly influenced its dominant knowledge-use practices.

These features of TekMar's pregraft context (a strongly held science-technology identity, along with knowledge content and knowledge-use practices consistent with that identity), made for potential problems in executing TekMar's strategic transformation via knowledge grafting. These themes constituted the basis of power and control tensions between leaders (especially members of the Venture Management Group) and middle managers that could influence the outcome of the change effort.

II. Graft Initiatives

The second column in Figure 3 deals with actions taken to perform the graft. Into the context described above, the Venture Management Group introduced strategic initiatives that implied the need for TekMar to change its identity, the content of its organizational knowledge, and its knowledge-use practices. These initiatives reflected changes that senior leadership deemed to be essential for ensuring TekMar's survival. In particular, senior leaders believed that TekMar needed to engage in an *identity change initiative*—to transform itself from a technology-push to a market-pull organization. Members of the Venture Management Group did not specifically tout the need for an identity change, per se; rather, it was the other members of the senior leadership team who articulated this aspiration. Continued use of terms like "market-driven," "market-oriented," and "customer focused" underscored the belief that the desired new business development capabilities required changes in the way TekMar defined itself. As one senior vice president noted, "We want to be more market-pull, no question about it. We're still technology pushers for the most part. That's just the way our training is. We're trying to evolve past that though, but it's really hard." In its now-independent form, TekMar had to find new markets and customers, something its scientists and engineers had never done before. A senior vice president explained: "We had a market that was defined for us into the 1980s by the parent company. They provided the definition of the world into which we invented. Once you have that cut off from you, then you struggle with how you define who you are." TekMar's identity was deeply ingrained, however, so it was clear that marketing would only supplement, not supplant, the technology-based identity.

In a related vein, the Venture Management Group launched a *marketing/business development knowledge content initiative* intended to introduce more marketing and business development skill

and expertise into TekMar at both the organizational and business-unit levels. A midlevel planning manager noted that TekMar's history as a place where the scientists "just continue to work on their dissertations" without thinking about commercial applications was difficult to change. A senior vice president gave an example of the new marketing skills needed:

When I go to a customer now, I can't go in, open up my box and say, "Look at all the technology I have." . . . Now you have to do your homework in advance and say to a company, "We looked at your market share. How would you like to improve it? We have something that will do this, this, and this for it."

Closely associated with the new knowledge content initiative, the Venture Management Group installed a new *formal business development procedures initiative*, which required the use of structured market analysis procedures, new project funding guidelines, and structured knowledge creation, sharing, and reporting processes. Many senior executives acknowledged the need to enhance collaboration and knowledge transfer between business units and project teams, so these changes had a convincing logic to them. We should emphasize that this initiative had much more to do with sharing marketing and business knowledge than technological knowledge (although we found that these new procedures nonetheless affected existing science and technology work practices, because projects that engineer/scientists could work on were now decided by the Venture Management Group's business development criteria, rather than traditional scientific criteria). TekMar's leadership realized that they needed to offer cross-business-unit solutions to customers, which, in turn, augured even more sharing of business knowledge. The major issue for the Venture Management Group, however, was convincing TekMar's engineers and scientists to embrace (formal) ways of sharing and using the business and marketing knowledge they were developing in light of their long-valued knowledge-use practices.⁷ A senior vice president noted this: "The pure technology stuff is less of a problem; the other, more significant one is the transfer of business knowledge. And I'll tell you what, we've been singularly poor in that process." The new formal procedures also meant that new projects would now be approved only after persuasive demonstration of their market potential and that funding for projects would be com-

petitive. The procedures also implied "airing our dirty laundry" because they required sharing information about project successes and failures with other business units. The Venture Management Group became an internal customer of sorts, in that TekMar's scientists and engineers had to *defend* their project ideas to the group if they were to have any chance of obtaining substantial financial support. This newly imposed subservience to the Venture Management Group conflicted with the scientists' sense of who they were collectively—an organization of world-class researchers.

III. Graft Implications and Responses

In Figure 3, the third and fourth columns present implications of and responses to the graft.⁸ The CEO had given the Venture Management Group a clear mandate and the power to streamline venture creation and evaluation processes, which, as noted, they implemented by formalizing knowledge-use, -sharing, and -reporting procedures. The main implication of this initiative was a *pronounced change in knowledge-use practices* at TekMar. The net effect (apparently unintended) was a transfer of power and control to the business development executives and a de facto top-down approach to knowledge creation and sharing, run by the Venture Management Group. Some of these structured operating procedures for launching and evaluating projects (especially those that forced the use of a formal "pathway" for moving new technology ideas from inception to market) were particularly disruptive of the scientists' preferred practices. Although these procedures did indeed compel the engineers/scientists to develop a deeper understanding of market trends and consumer needs, they also forced TekMar's technology champions to now follow unfamiliar and unwelcome procedures to get approval and funding for their projects. Members of the Venture Management Group maintained that their focus on using formal mechanisms to impart business development knowledge throughout the organization was not meant to affect day-to-day (scientific) activities, but the imposed procedures were nonetheless quite unsettling.

Of perhaps most significance (and of most interest to us, as researchers), this dramatic change in work practices undermined members' sense of

⁷ We should note that over the course of our study, there were no discernable alterations in the scientific practices used at TekMar.

⁸ In this findings narrative, we have combined columns 3 and 4 of Figure 3 and articulated the findings row by row to more clearly associate specific implications of the knowledge graft with particular responses by TekMar's members.

their *TekMar* identity ("It's just not who we are" [i.e., doing things in the newly prescribed way]). *TekMar*'s members, therefore, responded by engaging in attempts to *preserve their current identity*. They saw the shift in emphasis toward ventures and away from basic technology research as a threat to a deeply held and strongly valued collective identity. Furthermore, they saw the putative change in identity as damaging to *TekMar*'s reputation and ability to attract the best scientists and, therefore, as weakening their ability to continue the kind of work for which the company was famous. One business-unit manager remarked that the strong move toward ventures had led to *TekMar*'s researchers vying for "small ideas," which in turn reduced the organization's capacity to generate ground-breaking innovations. As a response, they developed alternative practices in line with their threatened identity, while (in their eyes) still meeting *TekMar*'s strategic needs (usually by adapting technology solutions across market domains, which we explicate below).

Interestingly, however, despite their preference for previous (identity-related) practices, the middle-level managers and some senior managers (who had been researchers themselves) expressed a clear *openness to marketing and business development knowledge content*. They readily accepted the logic and need for adding marketing and business development knowledge and readily incorporated such knowledge into their thinking about how their technological inventions might be developed for commercial application. Their attitude toward the addition of knowledge content, then, was one of acceptance, even if their actions in applying new knowledge content were not. We should also note, however, that their sense of "who we are" did not come across as problematic in the face of the new initiative to acquire new (business development) knowledge. Yet identity-related issues nonetheless surfaced whenever our informants felt that the new knowledge initiatives might affect *how* they used the knowledge they were expected to acquire. For example, a senior vice president remarked,

If technical people come to believe that all that matters is going to see customers, bringing in contracts, and doing things by the book, etc., well, God forbid because . . . let me tell you, if that gets to be pervasive, the organization will have completely forgotten what it is about. Because it's only about one thing; it's about a very high level of technology . . . applied to good market problems. . . . [trailing off, almost *sotto voce*] If that isn't the be all and end all of the corporation, people will find someplace else to go.

This consistent observation led us to conclude that *TekMar*'s organizational identity was to some extent recursively related to the development of new knowledge by shaping, and being simultaneously shaped, by both the current knowledge content and perhaps to a greater extent, the knowledge-use practices in the organization.

Incompatibility of identity and practice at TekMar. The key issue, then, and the main implication of the knowledge graft, had to do with *current identity–new practice incompatibility*. *TekMar* technologists saw the imposed procedures as at odds with practices they associated with *TekMar*'s identity as a center for world-class engineers and scientists, and it was this incompatibility that was at the heart of the tension between the scientists/engineers and the Venture Management Group people. The scientists/engineers wanted to retain their existing practices and still try to satisfy the Venture Management Group's ends, which would leave control of their work in the hands of the scientists/engineers. The Venture Management Group doubted that such an approach could work and tightened the requirements for the scientists to follow the formal procedures. The Venture Management Group members also tended to coningle issues of control and knowledge use, especially in frequent references to dealing with the "controlled chaos" that the business units were proud of practicing. In response to a question about how they could spark a greater market orientation, the Venture Management Group head noted:

You've got to keep the lid on it [chaos]. You know, it's like . . . you might be really liberal with your kid, but he still has to come to you for his allowance or for the keys to the car, or whatever it is. There isn't much he can do. It is the parenting process. I mean, there is a certain point where you give people latitude, and then there is always something that you sort of hang on to, so they can't go too far.

On another occasion the Venture Management Group head was considerably blunter, noting, "I have the power to force the issue here. I can force these scientists to follow SOPs, otherwise they won't get funding. It's as simple as that. That's our way of controlling the chaos around here." The struggle over this essential issue was threatening the strategic change effort itself.

Although some members of senior leadership, and especially the Venture Management Group people, might not have recognized that their imposed procedures and formal reporting systems were altering the ways that organizational knowledge was created, shared, and used in practice, the middle managers and some other senior managers

did. They expressed concern via warnings and insinuations that formalizing procedures did not augur well for the scientists, because such procedures hindered their creativity and their preference for more “chaotic” knowledge-use practices. Of fundamental importance was the finding that allusions to *organizational identity* almost always emerged when our respondents were expressing their concerns about threats to their current knowledge-use practices. As one senior vice president acknowledged near the end of the change effort, “The right of guys to be quirky is really the strength of the place. And you cannot, *cannot*, coerce them into a mode that they are just not willing to step into.” In essence, the forced formalization of procedures actually undermined the knowledge graft’s prospects for success because the existing practices were so closely associated with TekMar’s identity.

From the perspective of TekMar’s scientists/engineers, their de facto approach to addressing this issue was to embrace the addition of business development knowledge, but to *adapt current knowledge-use practices* to suit client demands. The COO, a scientist himself, noted,

What happened is, while the geeks [scientists] were developing a fairly sophisticated, unconventional skill achieving something, the system didn’t recognize what they were doing. . . . In fact, the geeks have worked out an approach to do this—they found a backdoor way to satisfy the demands of the market that was consistent with their geek way of doing things.

These “backdoor” practices were a demonstration of how the scientists were living out their acceptance of the need to be market-oriented in their work practices but applying business development knowledge consistent with their own identity and ways of doing.

To illustrate this key finding in more depth, we present a representative vignette that captures the situated use of knowledge that was part of the engineers’/scientists’ attempt to form new market opportunities and also symbolizes their understanding of being market oriented. This story, which was told to us by a scientist, represents the adaptation of practices from the pre-Venture Management Group era (that is, from the weaning period) that they believed satisfied TekMar’s strategic needs and the Venture Management Group’s wishes while enabling the engineers/scientists to preserve their preferred knowledge-use practices and their identity as they accommodated a threatening, un-TekMar-like power—the Venture Management Group.

I got together with a fellow here, Bob Johnson [a pseudonym], and I said Bob you know we need to do something different. And at that time I had a little project with a leading pharmaceutical company and I said, “Why don’t we focus on this whole biomedical area. We’re not doing anything in it. Maybe pieces here and there, but that would be interesting.” So we used to sit in my room and make phone calls to presidents of pharmaceutical companies. We would get invited to Johnson & Johnson, Eli Lilly, and we would go to these places and basically say, “We’re TekMar. We’re smart technologists.” We knew a lot about how to think differently. We tried to apply it to their industry. And as we were doing that we were actually kind of successful. A pharma company from the Northeast approached us with a technical problem that they were having with their bronchial inhaler. The problem they were facing was that the amount of active inhalant dispensed by the inhaler was erratic and often varied according to the lung power of the patient. They were not sure if this was a chemical problem or one that had to do with the mechanical delivery system of the inhaler. They came to us asking whether we could suggest some solutions. We took a look at it and realized that what these people really needed was a way to eliminate the propellant in the inhaler. One day back at TekMar, we were discussing this informally with some electronic engineers and one of them suggested that instead of using a propellant-driven mechanical system to dispense the active ingredients through the inhaler why don’t we electronically charge the active drug ingredient onto an inert carrier and thus eliminate the propellant completely. The electronic charging of powders on television tubes was a technology we already had. We just took that idea and applied it to bronchial inhalers and got a patent out of it. So as you see, we were doing electronics and now we’re doing pharmaceuticals. But the interesting thing was the method was the same. The method of looking at a process in a different market and asking where is it inefficient and how do you improve the efficiencies using technology. That approach served us well.

In this vignette, the scientist had simply been discussing his current conundrum with fellow scientists at coffee, and they drew the connection between a pharmaceutical problem and an existing solution they had developed in electronics. Yet the adaptation led to a wider application of existing (business development) knowledge use in practice to create a new market for the company. The vignette is an example of the ways that people went about attempting to understand their own experience in terms of who they were and what they did. It also shows how identity relates to the specific practices that undergird the use of knowledge. Note that this vignette does not primarily concern how these scientists knew about technology; it is how

they knew about markets and customers. They were using technological knowledge to know markets, in contrast to the Venture Management Group's mandate, "First know the market, then bring in technology."

We heard repeated variations on this theme that "we know the technology and can adapt our ways of knowing and doing to the market." The engineers and scientists approached potential client companies on the basis of their TekMar identity as "smart people who can solve customer business problems with technological solutions" ("That's who we are and what we do at TekMar"). The vignette also demonstrates how knowledge was translated or adapted for use in different market domains. Note also that this approach is not a technology-push approach. It instead represents technology adaptation or translation to a different market. The engineers and scientists reframed the inhaler problem as a market-relevant problem they could solve. They were trying to solve the necessary business development problems according to their own ways of thinking and doing as influenced by their TekMar identity.

Tracking practice change at TekMar. A fundamental dynamic in the case was TekMar's members' *ambivalence* toward the knowledge-grafting initiatives spearheaded by the Venture Management Group. On the one hand, there was near unanimity about the need for TekMar to embrace marketing and business development knowledge—that is, a desire to add new knowledge content to TekMar's repertoire. On the other hand, there was collective opposition to the formal business development procedures and market analysis methods that the Venture Management Group members instituted; this opposition produced a tension that manifested in sustained efforts by TekMar's members to preserve their science-based knowledge-use practices. These efforts led to localized adaptations of their traditional knowledge-use practices (as represented in the above vignette) to address the need for TekMar to become more market-oriented.

Because of our discovery of the importance of the identity-knowledge-practice nexus at TekMar, we conducted an additional comparative analysis to track changes (or an absence of changes) in TekMar's knowledge-use practices in three time periods: (1) the pregraft era, (2) the period when the knowledge graft was attempted, and (3) the post-graft period. This analysis revealed four prominent knowledge-use dimensions: (1) locus of knowledge use, (2) search for technology application opportunities, (3) customer interaction, and (4) knowledge sharing. These dimensions emerged from our comparing the areas the Venture Management Group

emphasized for the graft initiatives and our informants' detailed explanations of how they went about using and sharing knowledge day-to-day. The first column of Table 2 shows the four constituent dimensions of knowledge-use practices that emerged from our analysis.⁹ The second column describes the character of the scientific practices, which did not change throughout the graft period (as shown in column 5). The third column shows how the formal procedures of the Venture Management Group influenced each practice dimension, whereas the fourth column shows the resulting incompatibilities or conflicts with TekMar's current practices. The last column of the table shows how some of the scientists, as a part of their efforts to oppose the forced changes in their knowledge-use practices, developed their own distinct business development practices, based on modifications of their traditional scientific practices. These adapted practices, in the eyes of TekMar's members, not only addressed the overarching strategic goal of the organization's becoming more market- and customer-oriented, but simultaneously helped maintain their collective sense of who they were as an organization, while preserving their technical prowess. Table 2 thus acts as a "magnifying glass" showing how TekMar shifted incrementally from an insular technology organization to an organization trying to adapt its practices to become more of a technology/marketing organization in an effort to survive. We should note, however, that these adaptations did not reach a systemic level that would constitute a strategic transformation. There were local pockets of adaptation but no organization-level change.

A GROUNDED MODEL OF TEKMAR'S STRATEGIC CHANGE PROCESS

Strategic transformation is, almost by definition, so significant that it often entails a change to an organization's identity—or to the way knowledge is used in practices that might be associated with that identity. Our data in the study of TekMar's strategic transformation attempt strongly suggest that these two key features of an organization are actually closely related in such a way that a proposed or imposed change in one (e.g., practices) corresponds with destabilization in the other (e.g., identity) in ways that affect power relations within the company. In this section, we draw out the most important progressive and recursive relationships among these key concepts in a fashion that not only

⁹ We thank one of our anonymous reviewers for the suggestion to create a table to display this analysis.

TABLE 2
Tracking Changes in Knowledge-Use Practices at Tek-Mar

Before the Graft		During the Graft			After the Graft
Practice Dimensions	Scientific Practices	New Business Development Procedures	Sources of Deterrents to Change	Changes in Scientific Practices	Changes in Marketing and Business Development Practices
Locus of knowledge use	Aiming for scientific brilliance Refining scientific ideas	Driven by market and customer need Analysis of specific external prospects opportunities	The increased emphasis on market analysis undermined TekMar's collective identity based on technical excellence	No change	Adapting a technology over multiple market domains
Search for technology application opportunities	Chaotic and creative search processes. Scientific challenge as the primary search criterion	Formal market analysis procedures Projects approved only after demonstrating market potential	The formal procedures and processes weakened the creativity of scientists	No change	Serendipitous search over several industries Based on hunches about market trends and customer needs Hiring "jungle guides," or external market experts
Customer interaction	Negligible	Customers come first Need to tailor offerings to clients Need to be flexible and submissive to external funding sources such as venture capital firms	The key drivers of new ideas in TekMar were technical champions Focus on customers took attention away from big technical ideas	No change	Trying to motivate customers to buy into a technology's promise Impressing prospective customers with TekMar's technical prowess
Knowledge sharing	Ad hoc Based on mentoring and help giving Informal Independent and localized Experiential	Formal reporting procedures Company-wide dissemination of project successes and failures Open competition for project funding Forced sharing through formal meetings Revenue sharing precondition for knowledge sharing	Discomfort with revenue sharing Public discussion of failures What works in one business unit might not work in another	No change	Intermittent sharing between business units based on mutual benefits Informal help seeking from the Venture Management Group's members

helps explain TekMar's experience in theoretical terms, but also helps to explain that experience in terms that are broadly applicable to other strategic transformations as well.

As depicted in Figure 4, key concepts associated with virtually all organizations—identity, knowledge, knowledge-use practices, and power relations—become recursively interlinked as the members of an organization come to understand how proposed changes will affect their day-to-day lives. What we term “dual escalation cycles” form from the attempts to change (either implicitly or explicitly) an organization's identity and/or how knowledge is used in practice within the organization; each cycle builds on itself, as well as its companion cycle, to generate the type of deterrents to change observed both in this study and in previous organizational change research.

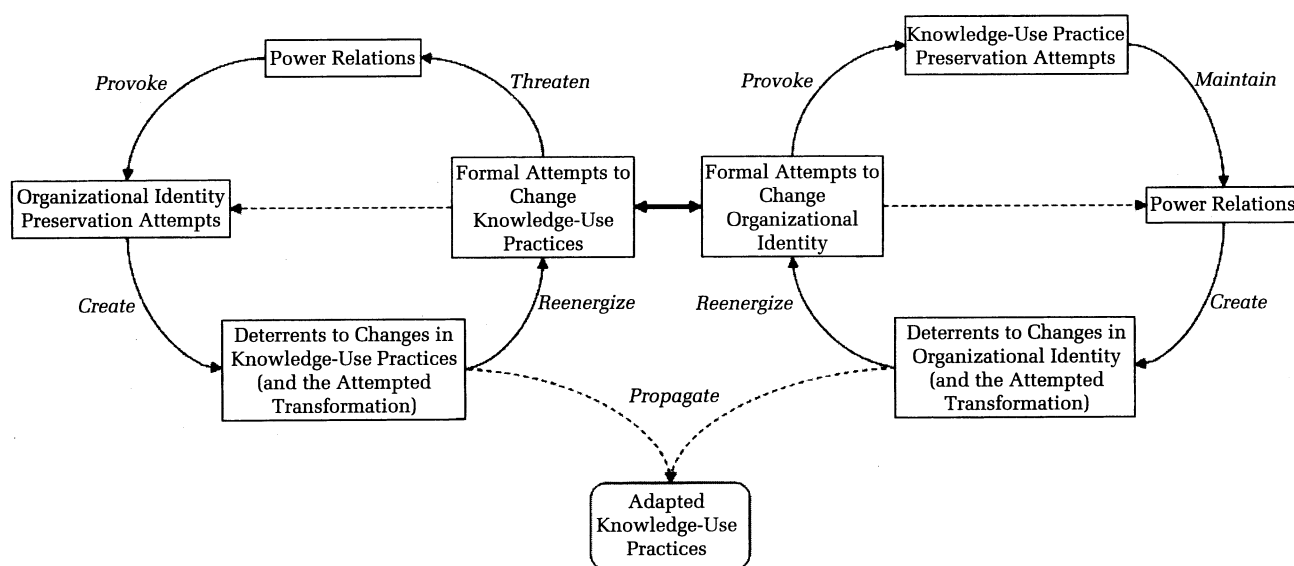
Figure 4 illustrates these recursive processes as revealed in TekMar's case. In the cycle depicted on the left, when deliberate attempts are made to change knowledge-use practices, power relations become threatened, not only concerning *who* controls knowledge, but also *how* it is controlled and used in the current change situation. Such threats to existing power relations provoke attempts to preserve current identity. Similarly, changes to knowledge-use practices (indicated by the straight dotted line in Figure 4) also provoke attempts to preserve current identity, as the knowledge-use practices provide the pragmatic substrate on which collective notions of identity form. Once identity is threatened and attempts to preserve it are enacted, a strong deterrent to changes in knowledge-use

practices and organizational transformation is created. This cycle then escalates when change agents respond to these deterrents with reenergized attempts to impose change, which again threaten power relations, thus continuing the cycle.

A related dynamic occurs, as is depicted on the right side of Figure 4. Deliberate attempts to change organizational identity provoke a desire to preserve how knowledge is used in practice, which in turn helps to maintain current power relations, creating further deterrents to changes in identity (and thus to the transformational change effort itself). The main difference in this escalation cycle is in the trigger for the proposed strategic change; here, it is explicit or implicit attempts to change the very definition of “who we are as an organization,” rather than attempts to change specific knowledge-use practices. Organizational identity frames, or creates meanings for, the knowledge-use practices of organization members. These collective practices form the basis for power relations among the organization members. Maintaining current power relations creates deterrents to identity change, which escalates the cycle when those deterrents are met with reenergized attempts at identity change. Changes to current organizational identity directly threaten existing power relations (indicated by the straight dotted line), further deterring change. The more general implication of beginning either cycle (i.e., imposing a change in knowledge-use practices or attempting a change in organizational identity), then, is to jeopardize the transformational change effort itself.

Although these two cycles have strong potential

FIGURE 4
The Dual Escalation Cycles of Strategic Change: A Grounded Theory Framework



to escalate and thus ensnare an organization in change-defeating cycles, the TekMar case ultimately revealed an alternative outcome: the development of adapted knowledge-use practices. TekMar's members constructed these adapted practices by melding aspects of their original knowledge-use practices with the features of the new procedures that made sense to them (see the vignette in our Findings section). Ultimately, this alternative path bypassed the escalating cycles (the curving dotted lines converging on "adapted knowledge-use practices" in Figure 4) and enabled TekMar's scientists and engineers to incorporate the new marketing and business development knowledge in their collective work practices while maintaining a sense of continuity for current organizational identity and successfully preserving their power bases. Such adaptive adjustments constitute a means for organizations with deeply ingrained practices associated with a deeply held identity to accomplish constructive, albeit not transformational or strategic, change.

We offer the following coda to the case: The situation at TekMar became increasingly precarious as we worked to complete this research project. Eventually, senior leadership (strongly supported by old guard, engineer/scientist business-unit managers) declared the knowledge graft a failure and suspended the attempt at strategic transformation. They eliminated the Venture Management Group because the group was now viewed not only as threatening TekMar's practices and identity, but also as superfluous and unnecessary. Key business development people in the business units soon quit. In the aftermath of our study, TekMar reverted to its earlier strategy of trying to succeed by pursuing its historical focus on developing and pushing cutting-edge technology into the market with variations on the business development practices they had improvised before and during the Venture Management Group's tenure. A recent consultation with a TekMar informant indicated that the failed transformation effort had major subsequent effects. Most of the executive staff had experienced turnover; venture attempts had declined; and a much broader range of market-oriented technology products and services had been introduced, including technology consulting. TekMar's current work practices, however, continue the adaptive practices that emerged during the transformational change effort.

DISCUSSION

An overarching revelation of our study is the importance of recursive interrelationships among

organizational knowledge, identity, and practice: the ways in which the organization members studied here used knowledge in their work practices affected and were affected by their collective notion of who they were as an organization. This study also suggests that the intersections among these major concepts have theoretical and empirical implications for how scholars conceptualize organizational knowledge, think about organizational identity, account for established organizational practices and, consequently, understand and study strategic and transformational change in organizations. Overall, the study demonstrates that knowledge is, in great part, grounded in its social context. In the following sections, we elaborate on these and other insights available from the study.

Organizational Change and Knowledge Use in Practice

All organizational change involves new knowledge in some form, whether it is technical knowledge, business knowledge, process knowledge, or some other type of knowledge introduced into the existing way work is accomplished. In itself, this statement is not a revelation, as the change literature has recognized and promoted new knowledge as a key element of organizational transformation (cf. Reger, Gustafson, Demarie, & Mullane, 1994). What is novel, however, is the recognition that it is not merely the presence of new content that is so important in bringing new knowledge into an organization, but rather, the effect new knowledge has on the collective knowledge-use practices of those most affected by a change. By focusing on the ways in which knowledge use is embedded in everyday practices and moving beyond a conception of knowledge as something possessed (a body of canons independent of organization members), our understanding of strategic change is enhanced as we direct our research less toward the role knowledge content plays in organizational change and more toward the ways that change recipients use both old and new knowledge to help create adaptive practices to deal with new environmental contingencies. An important theoretical implication is that knowledge becomes an *organizational* concept through its local manifestations in the collective practices of organization members. An important practical implication derives from recognizing that change initiation can focus either on content or process. For transformation to occur, an organization must address not only its knowledge content, but also its knowledge-use practices.

Our findings extend research into the more dynamic aspects of knowledge use, including the so-

cial factors that influence knowledge (Henderson & Clark, 1990), knowledge as a production context in its own right (Knorr-Cetina, 1999), and “knowing,” or the epistemic work done as a part of action or practice (Cook & Brown, 1999; Orlikowski, 2002). Our findings also suggest a new direction for research into knowledge transfer within and between organizations. Studies of knowledge transfer (cf. Gupta & Govindarajan, 2000; Simonin, 1999; Szulanski, 1996) have advanced understanding of *tangible* barriers to knowledge flows between organizational units (mostly by assessing measurable or objective knowledge characteristics). Our study underlines the importance of complementing this approach with a finer-grained look at the specific practices of knowledge use, to account for the sociocognitive and political undercurrents of the transfer process. Future research on knowledge transfer could benefit by identifying and tracking changes (or lack of changes) in knowledge-use practices throughout the transfer process. This approach could help researchers identify deterrents to knowledge transfer and, more importantly, improve understanding of the microprocesses by which knowledge is translated.

Deterrents to Change

The difficulty of accomplishing change is a perennial topic in organization study. Scholars often approach the study of change taking an essentially microlevel perspective, focusing on personal “resistance” to change by individuals (Dent & Goldberg, 1999; Piderit, 2000). Where more macrolevel approaches have been used, they have tended to focus on notions such as organizational inertia, emphasizing structural factors that act as barriers that make change difficult (Hannan & Freeman, 1984; Rumelt, 1995). On the basis of our findings, we have concluded that it is more useful to subsume both “resistance to change” and “barriers to change” under the more encompassing label of “deterrents to change.” Although both micro- and macrolevel deterrents certainly exist, our findings suggest that an alternative perspective might be more fruitful—that is, the view that the difficulties of organizational change emerge from collective ways of thinking and doing. One of our contributions, then, is in articulating how multilevel links affect social responses to change initiatives: deterrence to change occurs at the intersection of identity (“who we are”), knowledge (“what we know”), and practice (“what we do”). This insight is an obviously important issue for understanding change because of the influence language has over the direction of research (Locke & Golden-Biddle,

1997). That is, if one refers to “resistance” to change, one is likely to look for (individual-level) resistance; if one says “barriers” to change, one is likely to look for (macrolevel) barriers. Realizing that deterrents to change exist at multiple levels (including interstitial levels) can help future change researchers, as well as change managers, gain a more complete picture of the social responses to organizational change efforts.

The Social Context of Knowledge Use: Identity, Practice, and Power

The larger significance of our finding that useable knowledge was embedded in everyday practices is that it emphasizes the importance of the social context of knowledge use. The dual process cycles in Figure 4 clearly show the complexity of major organizational change. Within this complexity, however, lie a number of useful insights regarding (1) the role of organizational identity in knowledge-use practices, (2) the emergence of adaptive change from ostensibly failed transformation attempts, and (3) sources of power and control dynamics during organizational change.

Organizational identity. This study provides empirical grounds for expanding understanding of organizational identity in a way that moves beyond the usual cognitive focus and acknowledges a more important role for collective practices. Specifically, we found that in addition to providing a cognitive frame (i.e., creating meanings) for the work practices in which members engage, more importantly, organizational identity *inheres in work practices*—especially in the ways members use knowledge in those work practices. The upshot of this finding is a mandate for researchers to account not only for the cognitive dimension of identity (organization members’ beliefs about “who we are as an organization”), but also for a complementary dimension, one that involves “what we *do* as a collective.” Although some previous evidence of a relationship between identity and behavior exists (Stryker, 1968; Wells, 1978), our findings suggest a more significant shift in also including a practice-embedded aspect. This constitutes a marked change from simply saying that organizational identity has a behavioral component.

These findings help extend recent theoretical advances from identity research. For instance, whereas Ravasi and Schultz (2006) discussed the theoretical implications for organizational identity, image, and culture when identity is threatened from external sources, our study demonstrates that identity threats also come from internal sources, with associated implications for “who we are” and

“what we do” as an organization. Likewise, our study extends the findings of an examination of identity ambiguity during a spin-off by Corley and Gioia (2004). They noted that because organizational identity consists of both labels and their underlying meanings, identity transformations involve intensified sensegiving (cf. Gioia & Chittipeddi, 1991) via heightened communication and symbolic action by an organization’s leaders. TekMar’s experience illustrates, however, that because meanings are socially embedded in members’ practices, identity transformation must go beyond actions taken by leaders and also engage those whose daily practices will be most affected by a new identity. Finally, Fiol’s (2002) study of an organization attempting identity change illustrates how language plays a role and how phased changes at the individual level track with changes at the organizational level during identity transformation. Our focus on the practice aspects of identity transformation helps provide a more comprehensive picture both of how identity change occurs within the social milieu of an organization and what is at stake during these perilous endeavors.

Adaptive organizational change. The originally intended transformation undertaken by TekMar’s leaders was remarkable in both size and scope; nothing less than a profound conversion from a science-driven to a market-driven technology organization would have achieved the original goals of the program. This attempt failed; yet in its place appeared a different form of change that was easier to accept. Instead of becoming locked into the change-defeating cycles that constituted so much of the formal change attempt, adaptation occurred in knowledge-use practices. Because organizational identity and knowledge-use practices were recursively intertwined, however, our findings also suggest the possibility that members’ adaptive practices might generate some subtle identity adaptation. Even though labels might remain the same (Fiol, 2002), the adaptation of practices consistent with those labels can nonetheless facilitate changes in their meanings (Gioia et al., 2000), thus providing a way to “remain the same while changing” (cf. Gagliardi, 1986).

Power and organizational change. On first view, the TekMar case looks like a straightforward story of a power struggle for control over a company’s future. Looking more deeply, we see that it is more than that—that complex underlying processes manifest in such a power struggle. When considering processes involved in major organizational change, researchers tend to gravitate toward explanations focused on power because change seldom happens *without* power dynamics. The fact

that power becomes salient during change, however, tends to elide or distract from deeper explanations that have more to do with relationships among social processes. Thus, although power dynamics were present at TekMar, they did not emerge as the central explanation for the outcomes of the change attempt. In moving beyond power explanations and accounting for the recursive interrelationships among identity and knowledge use in practice, we show that the visible invocations of power are, in many ways, mainly a consequence of actions that disturb these social relationships. Power and control dynamics, therefore, might best be viewed as a symptom; their roots lie deeper.

We might also note that power need not necessarily be seen as based on people controlling others (Dahl, 1957; French & Raven, 1959); power also can be viewed as associated with and emerging from local organizational practices (Clegg, 1989; Townley, 1993). The way members of organizations enact practices affords them power to counter the power of others who might want to influence them to do something they otherwise would not do (Foucault, 1980). The link between power and the legitimization of knowledge-use practices is supported by the work of Goffman (1967) and Bourdieu (1977), who observed that power inheres in the practices and interactions in which people collectively engage. In our case, the Venture Management Group members were vested with formal power to bring about content changes in organizational knowledge. The power of the scientists, however, emerged from how they actually *used* marketing and business development knowledge in their practices (which corresponded with their scientific practices, as shown in Table 2). This collision of power bases led to conflict between power based on expertise and formal authority versus power based on knowledge-informed local practices. The scientists’ low-key adaptation of their practices was their own form of power use. Preferred knowledge-use practices prevailed, which is quite informative in itself, again showing how power and knowledge influence each other in the context of strategic change.

Limitations and Suggestions for Future Research

No empirical study is free of limitations, and our effort to understand and conceptualize the experiences of TekMar is no exception. We identify some limitations that can serve not only as qualifiers for the emergent theory, but also as signposts for future studies in this area.

Timing. We gained entry into TekMar within a year after the Venture Management Group had been installed. The six months in which we were en-

gaged as researchers was, therefore, the time when the repercussions of the graft initiatives were being most keenly felt throughout the organization. On the one hand, we believe that the timing of our entry was fortuitous in that it enabled us to witness the evolving dynamics that eventually led to the removal of the Venture Management Group and to attain a holistic perspective on the change effort—from the change agents as well as the change recipients. On the other hand, had we been able to study the pregraft period as well, we might have gained a deeper view of the emergence of the collective knowledge-use practices and their coemergence with the organization's identity and power relations. We instead had to rely on retrospective data about the period before the graft. Nonetheless, drawing on suggestions of Miller, Cardinal, and Glick (1997), we took steps to mitigate this limitation by interviewing multiple respondents on the same issues, using additional data sources (e.g., archives, presentations, and intranet exchanges), and observing key meetings and presentations (see also Lincoln & Guba, 1985).

Single case. We realize that by focusing on one type of organization—in this case, an R&D company—we might be limiting the transferability of our findings. We believe, however, that both the specific findings and especially the grounded theory are applicable to many high-tech organizations, which are facing ever-increasing pressures to stay on the technological cutting edge while simultaneously retaining “market-ability.” We also feel that our findings have some utility for organizations more broadly. The major concepts of knowledge, identity, practice, and power are obviously relevant to all organizations.

Conclusion: Changing Ways of Thinking about Change

At one level, we have conducted a grounded study of knowledge content and knowledge-use practices. Yet, to understand knowledge-use practices, we needed to understand how they were recursively involved with organizational identity. Subsequently, it became apparent that these intertwined relationships constituted the source of the tensions and power relations that helped undermine the transformational change effort itself. If understanding the complex interplay of identity, knowledge, and practice provides a basis for a better theoretical understanding of organizational change, it also provides a basis for a better understanding of some of the deterrents to change. Of this triad, knowledge (content) is probably the easiest to change. Valued knowledge-use practices and

deeply held identity are likely to be notably more difficult to change. The meta-message here is clear, however: the social context of knowledge and knowledge use matters. In that sense, it is not as important that we as change researchers understand the “deep structures” of organizational change (cf. Chomsky, 1957) so much as it is important that we come to understand the “deep processes” attending organizational change.

Although informative literatures address organizational identity, organizational knowledge, and a practice view of knowledge, they seldom do so in an integrative fashion. Empirical study of the intersection of these concepts yields a different theoretical understanding of strategic change, one that emphasizes some key underlying processes that can help explain why organizational transformation—undoubtedly one of the most complex and important endeavors in modern organizational life—is so very hard to accomplish.

REFERENCES

- Albert, S., & Whetten, D. 1985. Organizational identity. In L. L. Cummings & B. M. Staw (Eds.), *Research in organizational behavior*, vol. 7: 263–295. Greenwich, CT: JAI Press.
- Allen, T. J. 1977. *Managing the flow of technology: Technology transfer and the dissemination of technological information within the R&D organization*. Cambridge, MA: MIT Press.
- Allen, T. J., Tushman, M. L., & Lee, D. M. S. 1979. Technology transfer as a function of position in the spectrum from research through development to technical services. *Academy of Management Journal*, 22: 694–708.
- Bartunek, J. M. 1984. Changing interpretive schemes and organizational restructuring: The example of a religious order. *Administrative Science Quarterly*, 29: 355–372.
- Beer, M., & Nohria, N. 2000. Cracking the code of change. *Harvard Business Review*, 78(3): 133–141.
- Bourdieu, P. 1977. *Outline of a theory of practice*. Cambridge, U.K., and New York: Cambridge University Press.
- Brown, A. D., & Starkey, K. 2000. Organizational identity and learning: A psychodynamic perspective. *Academy of Management Review*, 25: 102–120.
- Brown, J. S., & Duguid, P. 2001. Knowledge and organization: A social-practice perspective. *Organization Science*, 12: 198–213.
- Burgelman, R. A. 2002. Strategy as vector and the inertia of coevolutionary lock-in. *Administrative Science Quarterly*, 47: 325–357.
- Chomsky, N. 1957. *Syntactic structures*. Gravenhage: Mouton.

- Clegg, S. 1989. *Frameworks of power*. London and Newbury Park, CA: Sage.
- Cook, S. D. N., & Brown, J. S. 1999. Bridging epistemologies: The generative dance between organizational knowledge and organizational knowing. *Organization Science*, 10: 381–400.
- Cook, S. D. N., & Yanow, D. 1993. Culture and organizational learning. *Journal of Management Inquiry*, 2: 373–390.
- Corley, K. G., & Gioia, D. A. 2004. Identity ambiguity and change in the wake of a corporate spin-off. *Administrative Science Quarterly*, 49: 173–208.
- Corley, K. G., Harquail, C., V., Pratt, M., G., Glynn, M. A., Fiol, C. M., & Hatch, M. J. 2006. Guiding organizational identity through aged adolescence. *Journal of Management Inquiry*, 15: 85–99.
- Daft, R. L. 1985. Why I recommended that your manuscript be rejected and what you can do about it. In L. L. Cummings & P. J. Frost (Eds.), *Publishing in the organizational sciences*: 193–209. Homewood, IL: Irwin.
- Dahl, R. 1957. The concept of power. *Behavioral Science*, 2: 201–215.
- Dent, E., B., & Goldberg, S. G. 1999. Challenging “resistance to change.” *Journal of Applied Behavioral Science*, 35: 25–41.
- Dewey, J. 1938. *Logic, the theory of inquiry*. New York: Holt.
- Dewey, J., & Bentley, A. F. 1949. *Knowing and the known*. Boston: Beacon Press.
- Dutton, J. E., & Dukerich, J. M. 1991. Keeping an eye on the mirror: Image and identity in organizational adaptation. *Academy of Management Journal*, 34: 517–554.
- Eisenhardt, K. M. 1989. Building theories from case study research. *Academy of Management Review*, 14: 532–550.
- Fiol, C. M. 2002. Capitalizing on paradox: The role of language in transforming organizational identities. *Organization Science*, 13: 653–669.
- Foucault, M. 1980. *Power/knowledge: Selected interviews and other writings, 1972–1977* [1st American ed.]. New York: Pantheon Books.
- French, J. R. P., & Raven, B. 1959. The bases of social power. In D. Cartwright (Ed.), *Studies in social power*: 150–167. Ann Arbor, MI: Institute for Social Research.
- Gagliardi, P. 1986. The creation and change of organizational cultures: A conceptual framework. *Organizational Studies*, 7: 117–134.
- Gioia, D. A., & Chittipeddi, K. 1991. Sensemaking and sensegiving in strategic change initiation. *Strategic Management Journal*, 12: 433–448.
- Gioia, D. A., Schultz, M., & Corley, K. G. 2000. Organizational identity, image, and adaptive instability. *Academy of Management Review*, 25: 63–81.
- Gioia, D. A., & Thomas, J. B. 1996. Identity, image, and issue interpretation: Sensemaking during strategic change in academia. *Administrative Science Quarterly*, 41: 370–403.
- Glaser, B., & Strauss, A. 1967. *Discovering grounded theory*. Chicago: Aldine.
- Glynn, M. A. 2000. When cymbals become symbols: Conflict over organizational identity within a symphony orchestra. *Organization Science*, 11: 285–298.
- Glynn, M. A., Lant, T. K., & Milliken, F. 1994. Learning about organizational learning: An umbrella of organizing processes. In C. Stubbart, J. Porac, & J. Meindl (Eds.), *Advances in managerial cognition and organizational information processing*, vol. 1: 43–84. Greenwich, CT: JAI Press.
- Goffman, E. 1967. *Interaction ritual: Essays on face-to-face behavior*. Garden City, NY: Anchor Books.
- Golden-Biddle, K., & Rao, H. 1997. Breaches in the boardroom: Organizational identity and conflicts of commitment in a nonprofit organization. *Organization Science*, 8: 593–621.
- Grant, R. M. 1996. Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17: 109–122.
- Gupta, A. K., & Govindarajan, V. 2000. Knowledge flows within multinational corporations. *Strategic Management Journal*, 21: 473–496.
- Hannan, M. T., & Freeman, J. 1984. Structural inertia and organizational change. *American Sociological Review*, 49: 149–164.
- Hansen, M. T. 1999. The search-transfer problem: The role of weak ties in sharing knowledge across organization subunits. *Administrative Science Quarterly*, 44: 82–111.
- Haveman, H. A. 1992. Between a rock and a hard place: Organizational change and performance under conditions of fundamental environmental transformation. *Administrative Science Quarterly*, 37: 48–75.
- Henderson, R. M., & Clark, K. B. 1990. Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35: 9–30.
- Huber, G. P. 1991. Organizational learning: The contributing processes and the literatures. *Organization Science*, 2: 88–115.
- Isabella, L. A. 1990. Evolving interpretations as a change unfolds: How managers construe key organizational events. *Academy of Management Journal*, 33: 7–41.
- James, W. 1963/1910. *Pragmatism and other essays*. New York: Washington Square Press.
- Kanter, R. M. 1990. *When giants learn to dance*. New York: Touchstone Book.

- Knorr-Cetina, K. 1999. *Epistemic cultures: How the sciences make knowledge*. Cambridge, MA: Harvard University Press.
- Kogut, B., & Zander, U. 1992. Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3: 383–397.
- Kogut, B., & Zander, U. 1996. What firms do? Coordination, identity, and learning. *Organization Science*, 7: 502–518.
- Lant, T. K. 1999. A situated learning perspective on the emergence of knowledge and identity in cognitive communities. In R. Garud & J. Porac (Eds.), *Advances in managerial cognition and organizational information*: 171–194. Greenwich, CT: JAI Press.
- Leonard-Barton, D. 1992. Core capabilities and core rigidities: A paradox in managing new product development. *Strategic Management Journal*, 13: 111–125.
- Levinthal, D., & March, J. 1993. The myopia of learning. *Strategic Management Journal*, 14: 94–112.
- Lincoln, Y. S., & Guba, E. G. 1985. *Naturalistic inquiry*. Beverly Hills, CA: Sage.
- Locke, K., & Golden-Biddle, K. 1997. Constructing opportunities for contribution: Structuring intertextual coherence and “problematizing” in organizational studies. *Academy of Management Journal*, 40: 1023–1062.
- Miller, C. C., Cardinal, L. B., & Glick, W. H. 1997. Retrospective reports in organizational Research: A reexamination of recent evidence. *Academy of Management Journal*, 40: 189–204.
- Nonaka, I., & Takeuchi, H. 1995. *The knowledge-creating company: How Japanese companies create the dynamics of innovation*. New York: Oxford University Press.
- Oliver, C. 1997. Sustainable competitive advantage: Combining institutional and resource-based views. *Strategic Management Journal*, 18: 697–713.
- Orlikowski, W. J. 2002. Knowing in practice: Enacting a collective capability in distributed organizing. *Organization Science*, 13: 249–273.
- Pentland, B. T. 1992. Organizing moves in software support hot lines. *Administrative Science Quarterly*, 37: 527–548.
- Pettigrew, A. M. 1985. *The awakening giant: Continuity and change in Imperial Chemical Industries*. Oxford, U.K.: Blackwell.
- Piderit, S. K. 2000. Rethinking resistance and recognizing ambivalence: A multidimensional view of attitudes toward an organizational change. *Academy of Management Review*, 25: 783–794.
- Ravasi, D., & Schultz, M. 2006. Responding to organizational identity threats: Exploring the role of organizational culture. *Academy of Management Journal*, 49: 433–458.
- Reger, R. K., Barney, J., Bunderson, S., Foreman, P., Gustafson, L. T., Huff, A. S., Martens, L., Sarason, Y., & Stimpert, L. 1998. A strategy conversation on the topic of organizational identity. In D. A. Whetten & P. Godfrey (Eds.), *Identity in organizations: Building theory through conversations*: 99–168. Thousand Oaks, CA: Sage.
- Reger, R. K., Gustafson, L. T., Demarie, S. M., & Mullane, J. V. 1994. Reframing the organization: Why implementing total quality is easier said than done. *Academy of Management Review*, 19: 565–584.
- Rumelt, R. P. 1995. Inertia and transformation. In C. A. Montgomery (Ed.), *Resource-based and evolutionary theories of the firm: Towards a synthesis*: 101–132. Boston: Kluwer.
- Sastry, M. A. 1997. Problems and paradoxes in a model of punctuated organizational change. *Administrative Science Quarterly*, 42: 237–275.
- Selznick, P. 1957. *Leadership in administration: A sociological interpretation*. New York: Harper & Row.
- Siggelkow, N. 2002. Evolution toward fit. *Administrative Science Quarterly*, 47: 125–159.
- Simonin, B. L. 1999. Ambiguity and the process of knowledge transfer in strategic alliances. *Strategic Management Journal*, 20: 595–623.
- Staw, B. M., Sandelands, L. E., & Dutton, J. E. 1981. Threat-rigidity effects in organizational behavior: A multilevel analysis. *Administrative Science Quarterly*, 26: 501–524.
- Stephenson, W. 1953. *The study of behavior: Q-technique and its methodology*. Chicago: University of Chicago Press.
- Strauss, A., & Corbin, J. 1990. *Basics of qualitative research*. Newbury Park, CA: Sage.
- Stryker, S. 1968. Identity salience and role performance: The relevance of symbolic interaction theory for family research. *Journal of Marriage and Family*, 30: 558–564.
- Szulanski, G. 1996. Exploring internal stickiness: Impediments to the transfer of best practice within the firm. *Strategic Management Journal*, 17: 27–44.
- Townley, B. 1993. Foucault, power/knowledge, and its relevance for human resource management. *Academy of Management Review*, 18: 518–545.
- Van Maanen, J. 1979. The fact of fiction in organizational ethnography. *Administrative Science Quarterly*, 24: 539–550.
- Van Maanen, J. 1988. *Tales of the field: On writing ethnography*. Chicago: University of Chicago Press.
- Wells, L. E. 1978. Theories of deviance and the self concept. *Social Psychology*, 41: 189–204.
- Yin, R. K. 2003. *Case study research: Design and methods* (3rd ed.). Thousand Oaks, CA: Sage.
- Zander, U., & Kogut, B. 1995. Knowledge and the speed

of the transfer and imitation of organizational capabilities: An empirical test. *Organization Science*, 6: 76–92.



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